M.Tech · Fundamentals of Artificial Intelligence

Mini Project Report

AI-Based University Enquiry Chatbot

using Knowledge Representation and Natural Language Processing

Project topic No. 29 · Suggested technique: Knowledge Representation + NLP

Submitted by

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Solution files submitted with this report: University_Enquiry_Chatbot.ipynb (solution notebook), chatbot/ (Python package), chat_cli.py (terminal chatbot), app_streamlit.py (web demo), data/ (knowledge base, rules, test sets).

Abstract

This project builds an enquiry chatbot for a fictional university. The chatbot answers questions about programmes, fees, eligibility, admissions, hostel, scholarships and placements. It combines a frame-based knowledge representation (a semantic network with inheritance), forward-chaining production rules for personalised reasoning, a natural-language processing pipeline, a dialogue manager and a vector-store retrieval fallback. On a labelled test set of 105 queries the hybrid chatbot answered 92.4% correctly (intent macro-F1 0.93), against 38.1% for an NLP + vector-retrieval chatbot and 31.4% for a keyword FAQ-matching chatbot built on the same knowledge. On a held-out set evaluated once after development it reached 80.0%. Every answer is produced in a few milliseconds (mean 1.4 ms) on a laptop CPU, works offline and comes with an explanation trace.

Keywords: knowledge representation, frames, semantic network, forward chaining, production rules, natural language processing, chatbot, vector store

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1. Problem Definition

Every admission season a university receives a very large number of repetitive enquiries by phone, e-mail and at the front desk, such as "What is the fee for B.Tech CSE?", "Am I eligible for M.Tech AI with 6.8 CGPA?", "When is the last date to apply?" and "Is there a hostel for girls?". Staff spend a lot of time answering the same questions. Answers are not available outside office hours and are sometimes inconsistent between staff members.

The objective is an intelligent enquiry chatbot that:

- understands free-form English questions, including paraphrases, abbreviations (B.E, M.Tech, CGPA, plus two) and spelling mistakes;
- answers factual questions from a structured knowledge base of the university;
- reasons about personal questions, such as eligibility for a programme or the fee after scholarships, and explains every decision;
- follows the conversation ("How long is it?" after a question about M.Tech AI) and asks for missing details;
- answers free-text FAQs (ragging, Wi-Fi, refunds) through semantic retrieval, and politely refuses out-of-scope questions.

Scope and assumptions: the knowledge base describes a fictional institution, "Greenfield University", with 9 programmes (B.Tech CSE, AI & DS, ECE and Mechanical; M.Tech AI and Data Science; MBA; MCA; Ph.D.), 6 departments, 5 facilities, the admission process and the scholarship policy. All data is synthetic and created only for this project. The chatbot is text-based, works in English and runs fully offline.

2. AI Problem Formulation

The chatbot is modelled as a knowledge-based agent. It perceives a sentence, updates its internal state (the dialogue context) and acts by choosing the answer that its knowledge base and inference rules support.

2.1 PEAS description

PEAS element	Description
Performance measure	Answer accuracy, intent precision / recall / F1, correct refusal of out-of-scope questions, response time, explainability
Environment	Prospective students and parents typing questions; the university knowledge (frames, rules, FAQ passages)
Actuators	Text answers, clarification questions, explanation traces
Sensors	The user's typed sentence

2.2 Properties of the task environment

Property	Value	Reason
Observability	Partially observable	The user's intention and personal facts are hidden and must be inferred from text
Determinism	Stochastic	The same need can be phrased in many different ways
Episodes	Sequential	Follow-up questions depend on earlier turns
Dynamics	Static	The knowledge does not change during a conversation
State space	Discrete	Finite sets of intents, entities and facts

Agents	Single agent	One chatbot serves one user per session

2.3 Formal formulation

- **Knowledge base (KB):** a set of frames F = {University, Program, UGProgram, PGProgram, btech_cse, ..., hostel, ...} linked by is_a edges. Slot lookup uses inheritance: value(f, s) = f.s if the frame stores slot s, otherwise value(parent(f), s).
- **Percept to query mapping:** the NLP pipeline maps a sentence x to a structured query q = (intent, entities), where intent belongs to 18 intents (program_fee, eligibility, hostel, ...) and the entities are programme, qualification, percentage, CGPA, entrance exam, category, subjects and so on.
- **Working memory (WM):** for a personal question, WM = facts about the programme (from the frames) plus facts about the applicant (from the sentence).
- **Operators (rules):** production rules of the form IF c1 AND c2 AND ... THEN add/modify fact (16 eligibility rules and 10 scholarship rules).
- **Inference and goal test:** forward chaining fires every applicable rule once (refraction) until a goal fact (academic_eligible, net_fee) is derived or no rule can fire.
- **Fallback:** if no intent is confident, the query is embedded as a vector and the closest FAQ passage is returned when its cosine similarity is above a threshold; otherwise the agent replies "out of scope".

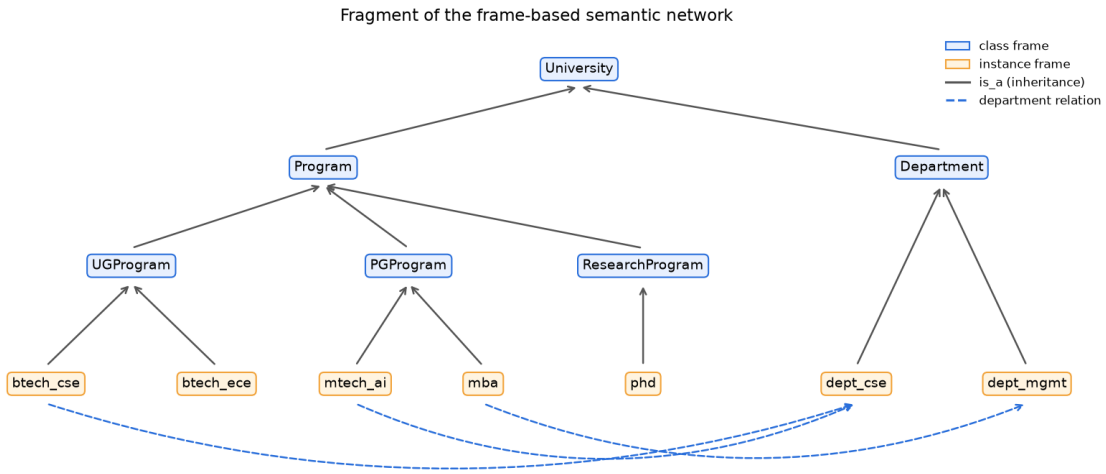


Figure 1. Fragment of the frame-based semantic network (is_a inheritance and department relations).

3. Selection and Explanation of the AI Technique

The topic asks for Knowledge Representation + NLP. The final design combines five classical AI components. Each was chosen for a specific reason:

Component	Technique used	Why it was chosen
Knowledge representation	Frames with is_a inheritance, which also form a semantic network of (subject, predicate, object) triples	University data is naturally hierarchical (Programme > UG/PG > B.Tech CSE). Inheritance stores duration, documents and deadlines once per class, so the KB is small and consistent.
Reasoning	Production rules + forward chaining with refraction and an explanation trace	Eligibility and scholarship policies are IF-THEN rules. Forward chaining is data-driven: it starts from the applicant's facts and derives the conclusion. Each fired rule becomes a sentence of explanation.
NLP	Normalisation, tokenisation, spelling	Transparent and explainable; needs only 109 example

	correction (edit-distance similarity), synonym mapping, stop-word removal, stemming, regex entity extraction, hybrid intent classifier	sentences; runs offline without a GPU; handles Indian academic abbreviations (B.E, M.Tech, CGPA, plus two).
Retrieval fallback	Vector store: TF-IDF word + character n-gram embeddings, cosine similarity, top-k search	Free-text FAQ passages (ragging, Wi-Fi, refunds) are not worth modelling as frames. The embedder is pluggable, so a neural sentence embedding can be used without code changes.
Dialogue management	Context memory + slot filling	Supports follow-up questions and asks for missing facts (marks, qualification) before running the rules.

3.1 Hybrid intent classifier

For each intent i the classifier computes score(i) = 0.6 x max cosine similarity between the query and the intent's example sentences (TF-IDF vectors of the pre-processed text) + 0.4 x keyword evidence (1 if a high-precision trigger word of the intent is present). The best intent wins. If even the best score is below 0.30, the query is passed to the vector store or answered as out of scope. A domain-relevance gate rejects questions that contain no word known to the university domain, such as "tell me a joke".

3.2 Forward chaining with explanations

For the question "I did B.E ECE with 6.8 CGPA and qualified GATE - am I eligible for M.Tech AI?" the working memory starts with the programme facts from the frame (minimum 60%, accepted qualifications, entrance exams) and the applicant facts from NLP (qualification = btech_ece, cgpa = 6.8, exam = gate). The engine then fires:

```
[R1]  Converted CGPA 6.8 to 64.6% using the standard formula (CGPA x 9.5).
[R2]  The minimum required aggregate for M.Tech Artificial Intelligence is 60%.
[R4]  Your qualification (B.Tech ECE) is accepted for M.Tech Artificial Intelligence.
[R6]  Your score 64.6% meets the cut-off of 60%.
[R11] There is no specific subject requirement for M.Tech Artificial Intelligence.
[R12] Conclusion: you satisfy the academic eligibility for M.Tech Artificial Intelligence.
[R14] Your GATE score is accepted for admission.
[R16] Because you qualified GATE you are eligible for a stipend: Rs. 12,400 per month.
```

3.3 Why not a pure machine-learning or LLM chatbot?

For admissions, answers must be exactly correct (fees, cut-offs), consistent and explainable. A knowledge-based system guarantees that every answer can be traced to a frame slot or a rule. A large language model may produce wrong numbers and cannot easily justify an eligibility decision. It would also need internet access or a GPU. The cost of the knowledge-based approach is manual knowledge engineering, which is discussed in Section 8.

4. Python Implementation

The solution is a small, UI-independent Python 3.12 package. The same engine is used by the solution notebook, the terminal chatbot (chat_cli.py) and the web demo (app_streamlit.py). Libraries: numpy, pandas, scikit-learn (TF-IDF, metrics), matplotlib, and streamlit for the web demo. The NLP and reasoning components use only the Python standard library.

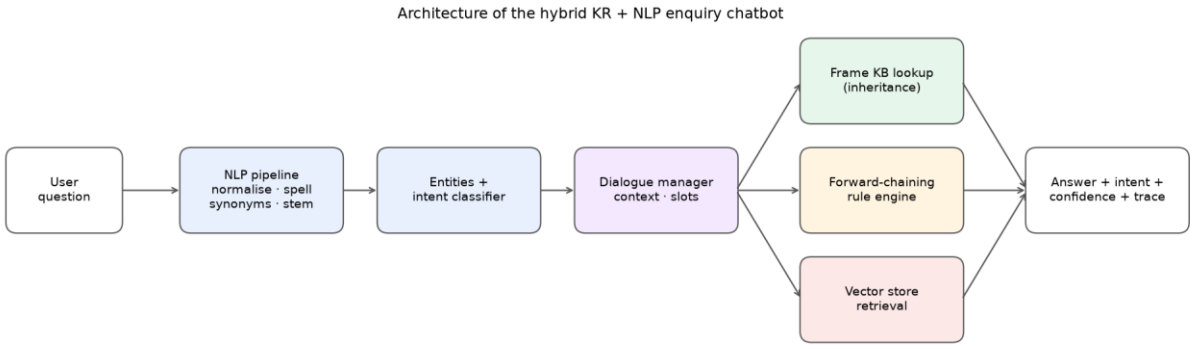


Figure 2. Architecture of the hybrid KR + NLP enquiry chatbot.

Module	Responsibility
knowledge_base.py	Frames, is_a inheritance (get, get_with_source, resolved), semantic-network triples and pattern queries
nlp.py	Normalisation of abbreviations, tokenisation, spelling correction, synonyms, stop words, stemming, entity extraction
intent_classifier.py	Hybrid TF-IDF similarity + keyword intent classifier with a confidence threshold
reasoning.py	Generic forward-chaining production-rule engine (conditions, actions, refraction, explanations)
vector_store.py	In-memory vector store with a pluggable embedder (TF-IDF default; sentence-transformers optional)
answers.py / faq_builder.py	Natural-language answer templates filled from frames; FAQ corpus for retrieval and the baselines
engine.py	UniversityChatbot: dialogue manager (context, slot filling), routing to KB / rules / vector store, trace
baselines.py / evaluation.py	Alternative approaches for comparison; test harness and metrics
chat_cli.py / app_streamlit.py	Terminal chatbot (with /why explanations) and web demo
data/*.json, data/*.csv	Knowledge base (29 frames), rules (26), intents (18), free-text FAQs (15), test sets

Processing steps for every message (UniversityChatbot.respond):

1. NLP pre-processing, entity extraction and intent classification.
2. If a task is pending (e.g. eligibility waiting for marks), the new facts fill the slot and the task continues.
3. Personal questions with applicant facts are sent to the forward-chaining engine (eligibility or scholarship rules).
4. The dialogue context resolves references ("it") and elliptical follow-ups ("What about MBA?").
5. The domain gate and the vector store handle unknown or low-confidence questions.
6. Otherwise the answer is filled from the frame KB. The response carries the text, intent, confidence, answer source and a full trace.

5. Dataset and Test-Case Preparation

There is no public dataset for a fictional university, so the knowledge and the test cases were prepared for this project:

- **Knowledge sources:** 29 frames, 26 production rules, 18 intents with 109 example sentences, and 90 FAQ passages (75 generated automatically from the frames and 15 free-text passages).
- **Main test set:** 105 hand-written queries, deliberately different from the intent examples, in 8 categories that test different abilities. Each query is labelled with the expected intent and the

expected answer key (e.g. program_fee:btech_ece, eligibility:mtech_ai:yes). An answer counts as correct only when the key matches exactly, so for reasoning questions the eligibility decision itself must be right.

- **Held-out test set:** 25 further queries written after development finished and evaluated only once, without any change to the system. This gives an unbiased estimate, because the main test set was also used to find bugs.

Category	Main set	Held-out	What it tests
factual	20	2	Direct factual questions
paraphrase	20	6	Same needs, different wording and synonyms
typo	10	3	Spelling mistakes ("hostle feez", "admision procedure")
reasoning	16	5	Personal eligibility / fee-after-scholarship decisions
multi_turn	14	3	5 conversations with follow-ups and slot filling
unstructured_faq	12	3	Free-text FAQ topics not modelled as frames
out_of_scope	10	3	Unrelated questions that must be refused
smalltalk	3	0	Greeting, thanks, goodbye
Total	105	25	

6. Performance Evaluation

Metrics used:

- **Answer accuracy:** share of queries whose answer key matches the expected key (main metric).
- **Intent accuracy and macro precision / recall / F1:** how well the question type is understood.
- **Out-of-scope precision / recall:** whether the chatbot refuses the right questions and only those.
- **Recall@k of the vector store:** whether the right free-text passage is among the top-k results.
- **Latency:** mean response time per query.

Metric	Hybrid KR + NLP
Answer accuracy	0.924
Intent accuracy	0.924
Intent macro precision	0.931
Intent macro recall	0.953
Intent macro F1	0.934
Out-of-scope precision	0.889
Out-of-scope recall	0.800
Mean latency (ms)	1.37
Vector store Recall@1 / Recall@3	0.92 / 1.00
Latency 95th percentile / max (ms)	1.86 / 2.72

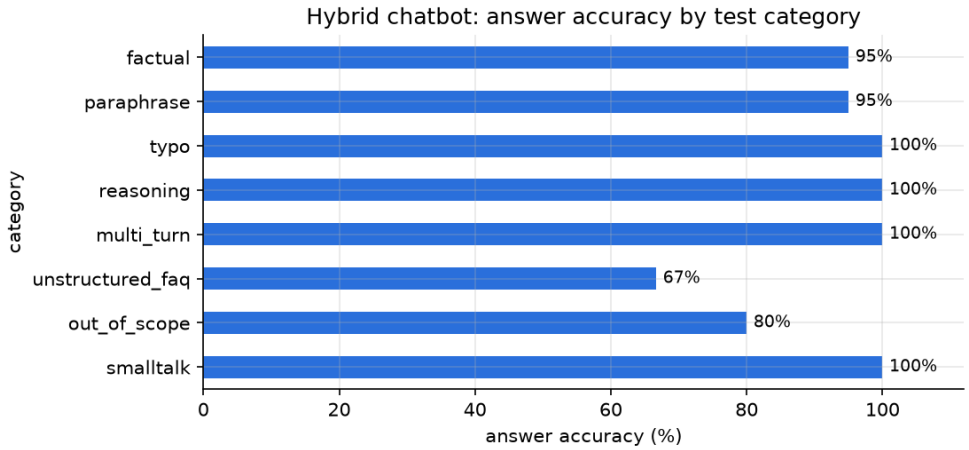


Figure 3. Answer accuracy of the hybrid chatbot by test category.

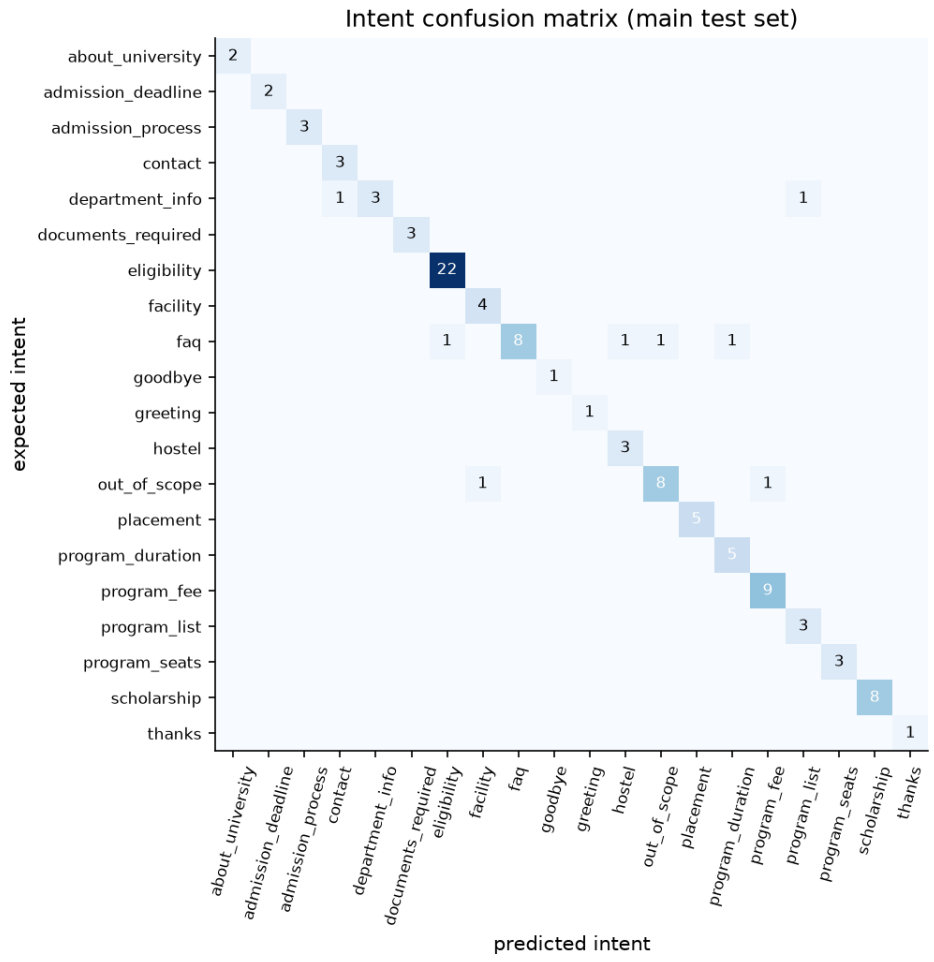


Figure 4. Intent confusion matrix on the main test set.

6.1 Error analysis

ID	Query	Predicted	Cause
F18	Which department offers MCA?	program_list:all	"offers" sent the query to the programme list instead of the department relation
P18	Where is the computer science department located?	contact	the word "located" triggered the contact intent
U02	Do you have wifi in the hostel?	hostel	"hostel" is a confident KB intent, so the Wi-Fi FAQ was never consulted

U03	Can diploma holders get lateral entry?	eligibility:ask_program	"diploma" was read as an applicant qualification, so the bot asked which programme
U08	Do we have to wear a uniform?	fallback	"uniform" has no lexical overlap with "dress code" (a limit of lexical embeddings)
U09	Can I study abroad for a semester?	program_duration:all	"semester" pulled the query towards the duration intent
O03	Can you book a movie ticket for me?	facility:all	"book" matched the library facility
O07	What is the share price of Infosys?	program_fee:all	"price" is a synonym of fee in the domain vocabulary

7. Comparison with Alternative Approaches

Two alternative chatbots were built on exactly the same knowledge: the same 90 FAQ question-answer pairs, with identical answer wording. Only the approach differs:

System	Description
Keyword FAQ matching (classic baseline)	Raw lower-cased text, word TF-IDF and cosine similarity to FAQ questions. No NLP, no KR, no reasoning, no memory.
NLP + vector retrieval	The same NLP normalisation and vector store as the proposed system, but every answer is retrieved from the FAQ corpus. No frames, rules or dialogue state.
Hybrid KR + NLP (proposed)	Frames + forward chaining + NLP + dialogue manager + vector-store fallback.

Metric	Keyword FAQ matching	NLP + vector retrieval	Hybrid KR + NLP (proposed)
Answer accuracy	0.314	0.381	**0.924**
Intent accuracy	0.419	0.552	**0.924**
Intent macro precision	0.482	0.578	**0.931**
Intent macro recall	0.432	0.502	**0.953**
Intent macro F1	0.388	0.490	**0.934**
Out-of-scope precision	0.077	0.167	**0.889**
Out-of-scope recall	0.100	0.400	**0.800**
Mean latency (ms)	0.30	0.65	**1.37**

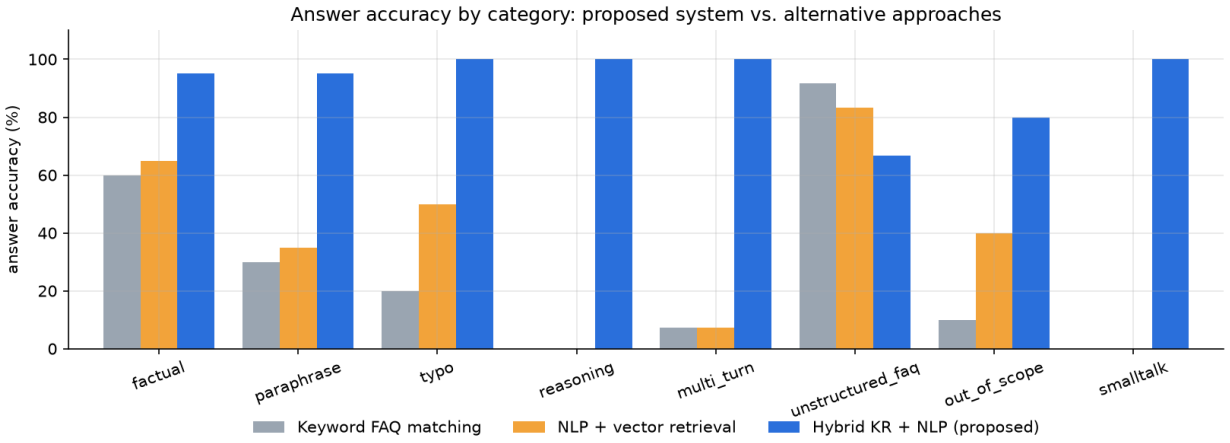


Figure 5. Answer accuracy by category: proposed system vs. alternative approaches.

7.1 Ablation study

To measure what each component contributes, one component was removed at a time:

Configuration	Overall	fact.	para.	typo	reas.	multi	FAQ	OOS
Full hybrid system	92.4%	95	95	100	100	100	67	80
without rule-based reasoning	72.4%	95	95	100	0	57	75	80
without dialogue context	83.8%	95	95	100	100	36	67	80
without spelling correction	86.7%	95	95	40	100	100	67	80
without vector-store fallback	84.8%	95	95	100	100	100	0	80

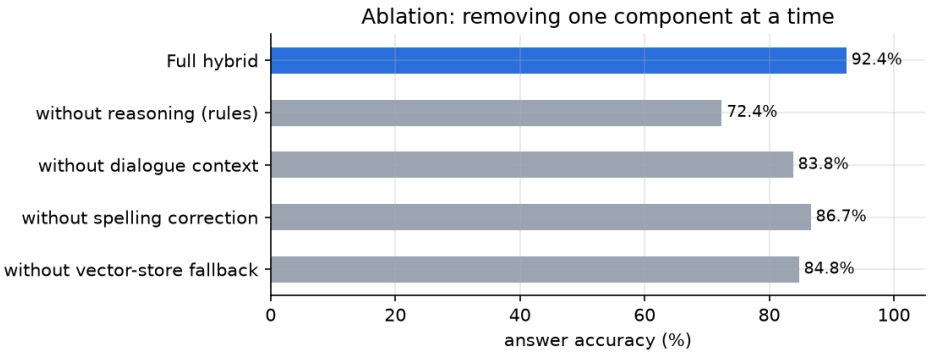


Figure 6. Ablation: answer accuracy when one component is removed.

7.2 Sensitivity to the confidence threshold

The intent-confidence threshold was fixed at 0.30 before testing. The sweep in Figure 7 shows that higher thresholds (0.35-0.55) give slightly higher accuracy and better out-of-scope recall on this test set. This was not adopted, because choosing a parameter on the test set would make the reported result optimistic. It should be validated on new data.

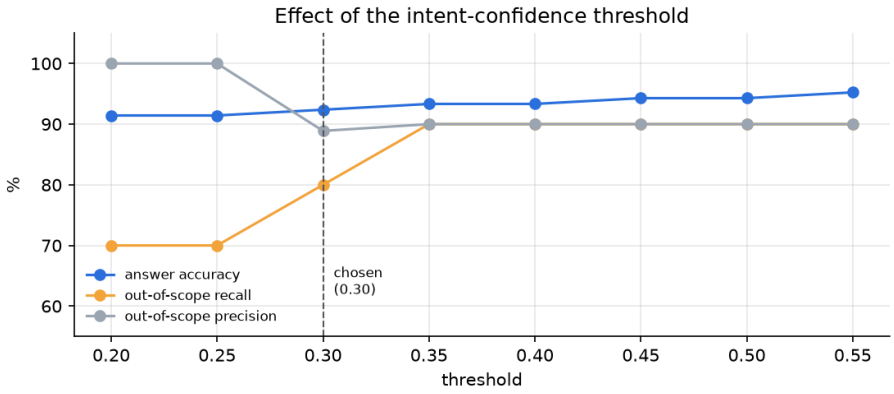


Figure 7. Effect of the intent-confidence threshold.

7.3 Held-out test set

Metric	Keyword FAQ matching	NLP + vector retrieval	Hybrid KR + NLP (proposed)
Answer accuracy	0.320	0.480	**0.800**
Intent accuracy	0.360	0.520	**0.800**
Intent macro F1	0.279	0.358	**0.670**
Out-of-scope recall	0.333	0.667	**0.333**

Note: while checking the held-out results, one label (H14, B.Sc Maths for M.Tech Data Science) was found to be wrong. The knowledge base accepts M.Sc Maths, not B.Sc, so the chatbot's "not eligible" answer was correct. The label was corrected; the system itself was not changed.

8. Results, Discussion and Conclusion

8.1 Results

- **Main test set:** on the 105-query main test set the hybrid chatbot answered 92.4% correctly (intent macro-F1 0.93), against 38.1% for NLP + vector retrieval and 31.4% for keyword FAQ matching.
- **Held-out set:** 80.0% for the hybrid chatbot, against 48.0% and 32.0%. The drop from the main test set is the expected optimism of a test set that was also used for debugging; the ranking of the three approaches is the same.
- **Speed:** mean 1.4 ms per answer on a laptop CPU, with no GPU and no internet.

8.2 Discussion and individual observations on the outputs

1. **Reasoning is what a retrieval bot cannot do.** Both alternatives scored 0% on reasoning questions and about 7% on multi-turn questions. They can return the general eligibility text, but not the decision for a specific applicant ("72% in B.E ECE, so eligible for M.Tech AI"). Removing the rule engine alone drops the hybrid from 92.4% to 72.4%, the largest single drop.
2. **Inheritance keeps the knowledge base small and consistent.** Duration, documents and deadlines are stored once per class (UG / PG / Research) and inherited by every programme. The trace shows where each value came from, e.g. mba.duration comes from the PGProgram frame.
3. **Explanations come for free.** Every fired rule produces a sentence ("[R3] A 5% relaxation applies for the OBC category, so the cut-off becomes 55%"), so a student can see why they are or are not eligible. A statistical chatbot does not give this.
4. **Small NLP steps matter.** Spelling correction raised typo accuracy from 40% to 100%, and normalising B.E / B.Tech / M.Tech / plus two / CGPA was essential for correct entity extraction.
5. **Dialogue context is necessary for natural conversation.** Without context memory, multi-turn accuracy falls from 100% to 36%. Users naturally ask "How long is it?" or answer a clarification with just "I have 61 percent".
6. **Structured and unstructured knowledge complement each other, but there is a trade-off.** Without the vector store, free-text FAQ accuracy is 0%; with it, 67% (Recall@3 = 1.00). On these questions the plain retrieval baselines score higher (83-92%), because the hybrid sometimes sends a question to a confident KB intent ("Wi-Fi in the hostel" goes to the hostel frame).
7. **Out-of-scope detection is the weakest part.** The domain gate catches clearly unrelated questions (recall 80.0%), but questions that share domain words ("share price of Infosys", "book a movie ticket") still get a university answer.

8.3 Limitations

- Knowledge must be engineered by hand; rules cover only eligibility and scholarship policies.
- TF-IDF embeddings are lexical, so pure synonyms ("in charge of" vs "HOD", "uniform" vs "dress code") are missed.
- English only; the knowledge base is synthetic; the test sets were written by the developer, not collected from real users.

8.4 Conclusion

A hybrid Knowledge Representation + NLP chatbot was designed, implemented and evaluated. Frames with inheritance hold the structured university knowledge. Forward-chaining production rules answer personalised eligibility and fee questions with step-by-step explanations. A lightweight NLP pipeline turns free-form questions into structured queries, a dialogue manager handles follow-ups and missing

details, and a vector store answers unstructured FAQs. The system clearly beats retrieval-only alternatives, especially on reasoning and multi-turn questions, while staying fast, explainable and offline.

Future work: replace the TF-IDF embedder with a neural sentence embedding (the Embedder interface already supports this), learn intent weights from real chat logs, add backward chaining for "what do I need to become eligible?" questions, support Tamil and Hindi, and connect the knowledge base to the live admissions database.

9. Demonstration of the Working Prototype

The prototype has two interfaces on top of the same engine. The terminal chatbot is part of the submission; the web version is for live demonstration.

9.1 Terminal chatbot (python chat_cli.py)

A real session with the terminal chatbot. The bracketed line after each answer shows intent, answer source and confidence; /why prints the reasoning trace.

```
Greenfield University - Enquiry Chatbot  (type /help for examples, /exit to quit)

You: What is the fee for M.Tech AI?
Bot: The tuition fee for M.Tech Artificial Intelligence is Rs. 1,20,000 per year. The total tuition for
2 years is about Rs. 2,40,000. A one-time application fee of Rs. 1,000 and a refundable caution deposit
of Rs. 10,000 are also payable. Hostel with mess costs Rs. 85,000 per year extra.
     [program_fee | knowledge_base | 0.91]
You: How long is it?
Bot: M.Tech Artificial Intelligence is a 2 years (4 semesters) full-time, on campus programme.
     [program_duration | knowledge_base | 0.79]
You: What about MBA?
Bot: MBA (Master of Business Administration) is a 2 years (4 semesters) full-time, on campus programme.
     [program_duration | knowledge_base | 0.23]
You: Am I eligible for M.Tech AI?
Bot: To check your eligibility for M.Tech Artificial Intelligence, please tell me your qualification
(e.g. 'B.Tech ECE', 'B.Sc', '12th') and your percentage or CGPA.
     [eligibility | dialogue | 1.00]
You: I did B.E ECE with 6.8 CGPA and I qualified GATE
Bot: Yes, you are eligible for M.Tech Artificial Intelligence. Converted CGPA 6.8 to 64.6% using the
standard formula (CGPA x 9.5). Your qualification (B.Tech ECE) is accepted for M.Tech Artificial
Intelligence. Your score 64.6% meets the cut-off of 60%. Your GATE score is accepted. You will also get
a GATE stipend of Rs. 12,400 per month for GATE-qualified students.
     [eligibility | rule_engine | 0.72]
You: /why
  Forward chaining fired: R1, R2, R4, R6, R11, R12, R14, R16
  [R1] Converted CGPA 6.8 to 64.6% using the standard formula (CGPA x 9.5).
  [R2] The minimum required aggregate for M.Tech Artificial Intelligence is 60%.
  [R4] Your qualification (B.Tech ECE) is accepted for M.Tech Artificial Intelligence.
  [R6] Your score 64.6% meets the cut-off of 60%.
  [R11] There is no specific subject requirement for M.Tech Artificial Intelligence.
  [R12] Conclusion: you satisfy the academic eligibility for M.Tech Artificial Intelligence.
  [R14] Your GATE score is accepted for admission.
  [R16] Because you qualified GATE you are eligible for a stipend: Rs. 12,400 per month for GATE-
qualified students.
You: I scored 96% in 12th and I am a state level athlete. What will my fee be for B.Tech CSE?
Bot: Fee estimate for B.Tech Computer Science and Engineering: tuition Rs. 1,50,000 per year. Merit
scholarship: 96.0% >= 95% gives a 50% tuition waiver. Sports quota (state/national level) adds a 25%
waiver. The total waiver is capped at 60%. Net tuition fee = Rs. 1,50,000 - 60% = Rs. 60,000 per year.
Your estimated tuition is Rs. 60,000 per year (scholarship 60%).
     [scholarship | rule_engine | 0.78]
You: hostle fees for girls?
Bot: Separate hostels for boys (1,200 beds) and girls (900 beds) inside the campus. Rooms: single,
double and triple sharing. The fee is Rs. 85,000 per year, which includes accommodation and mess (veg
and non-veg). AC rooms cost Rs. 20,000 extra per year. In-time 9:00 PM for all students.
```

```
     [hostel | knowledge_base | 0.88]
You: Is the campus ragging free?
Bot: Greenfield University has zero tolerance for ragging. An anti-ragging committee and squad monitor
the campus and hostels 24x7, and students can report incidents anonymously on the helpline 1800-180-
5522.
     [faq | vector_store | 0.77]
You: Who won the cricket match yesterday?
Bot: Sorry, I could not find an answer to that. I can help with programmes, fees, eligibility,
admissions, deadlines, documents, hostel, scholarships, placements and campus facilities. For anything
else please contact the admissions office at +91 44 2345 6789.
     [out_of_scope | fallback | 0.26]
```

9.2 Web demo (streamlit run app_streamlit.py)

The web demo runs locally at http://localhost:8501. The left panel holds sample questions, the centre shows the conversation, and the right panel explains each answer: intent, confidence, answer source, the rules that fired and the extracted entities.

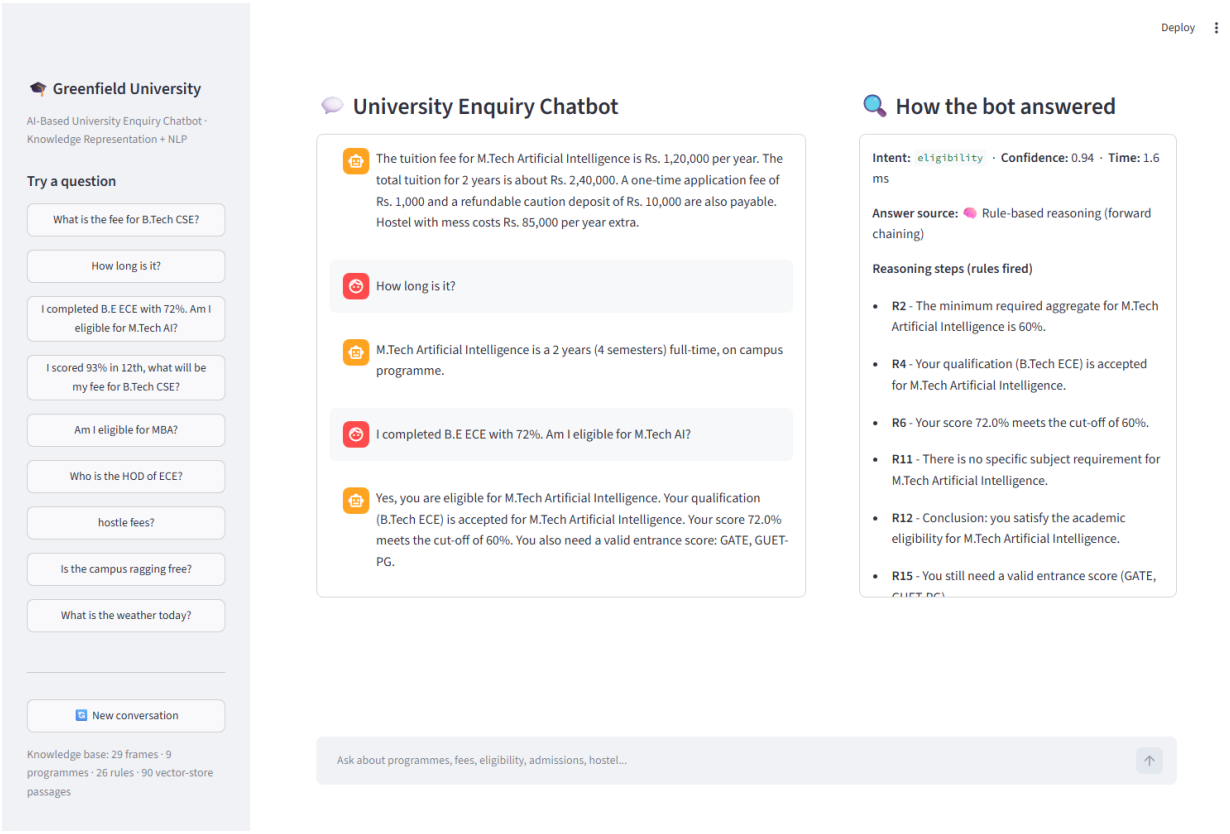


Figure 8. Web demo: follow-up question resolved from context, then an eligibility decision with the rules that fired.

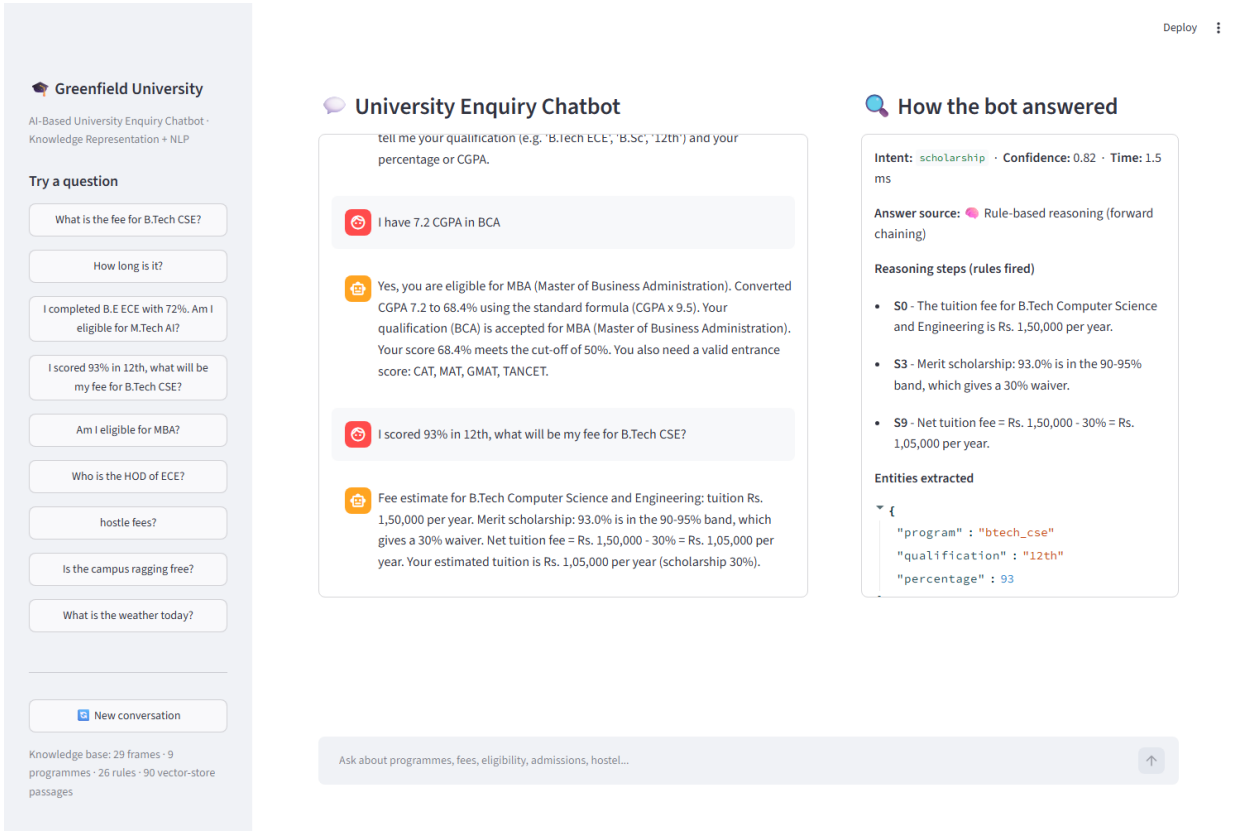


Figure 9. Web demo: slot filling (CGPA converted to percentage) and a fee calculation with merit scholarship.

Appendix A. Software Setup: Steps Followed

All software is free. It was installed for the current Windows user only, without administrator rights:

1. **Python 3.12.10:** winget install --id Python.Python.3.12 -e --scope user
2. **Visual Studio Code 1.141:** winget install --id Microsoft.VisualStudioCode -e --scope user
3. **VS Code extensions:** Python (ms-python.python) and Jupyter (ms-toolsai.jupyter), via code --install-extension
4. **Python libraries:** python -m pip install numpy pandas scikit-learn matplotlib python-docx jupyter ipykernel nbformat nbconvert streamlit
5. **PATH:** the Python and Python\Scripts folders were added to the user PATH so that python, pip and streamlit work in any terminal

Appendix B. How to Run the Solution

1. **Open the project:** in VS Code choose File > Open Folder and select the university_chatbot folder.
2. **Notebook:** open University_Enquiry_Chatbot.ipynb, select the Python 3.12 kernel (top-right) and click Run All.
3. **Terminal chatbot:** open Terminal > New Terminal and run: python chat_cli.py (commands: /why, /reset, /help, /exit)
4. **Web demo:** in the terminal run: streamlit run app_streamlit.py then open http://localhost:8501 (add ?demo=1 for a sample conversation).
5. **Re-run the evaluation:** the notebook recomputes every metric and figure from data/test_queries.csv.

References

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